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Finite-Horizon Q-Learning with Softmax: Early Rewards, Initialization, Path Dependence, and Under-Sampling

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Background: finite-horizon Q-learning and softmax exploration

Finite-horizon Q-learning learns action values from episode experience, starting from initial Q-values and updating them until a terminal state is reached. Softmax exploration turns those current Q-values into action probabilities, with a temperature parameter controlling how random or greedy behavior is. (7 sources)

In finite-horizon or episodic Q-learning, the agent starts with initialized Q-values, takes actions, observes rewards and next states, and updates Q-values from that experience until the episode ends at a terminal state. This basic setup makes current value estimates depend both on observed outcomes and on the initial values chosen before learning starts. (Nocera et al., 2014) (Zhang, 2024) (Higashi, 2025)

Q-learning itself does not force one specific exploration rule, so practice often adds a stochastic action policy to balance exploration and exploitation. A common choice is softmax, also called Boltzmann exploration, which converts current action values into a probability distribution over actions rather than always taking the current best-looking action. (Moriyama et al., 2007) (Nocera et al., 2014) (Diaz et al., 2025) (Zhang, 2024)

Under softmax, the temperature parameter controls randomness: high temperature makes behavior closer to random exploration, while low temperature makes the policy concentrate more on actions with larger current Q-values. Some papers express this with temperature, and others with inverse temperature or reward sensitivity, but the role is the same: it sets how strongly current value differences shape choice. (Nocera et al., 2014) (Moriyama et al., 2007) (Higashi, 2025)

This matters because the standard convergence story for Q-learning is asymptotic and depends on strong conditions: the environment should be Markov, learning rates should decay properly, and the agent must visit all states and take all actions infinitely often. Softmax is often used precisely because always picking current best actions can trap learning in limited parts of the state space, but the guarantee still rests on infinite exploration conditions rather than what must happen in a short finite run. (Moriyama et al., 2007)

More broadly, when learning is done with finite rollouts or finite horizons, estimates can differ systematically from the infinite-horizon quantities one may ultimately care about, so finite-time behavior is not just a small-sample version of the asymptotic theory. That gap is important for understanding later issues like early reward shocks, initialization effects, and practical under-sampling under softmax. (Zhang et al., 2019) (Baxter et al., 2001)

Evidence

(Nocera et al., 2014)

The role of intrinsic motivations in attention allocation and shifting

"In QL techniques, the Q-values are estimated through the agent experience after being initialized to arbitrary numbers"

".Different exploration policies can be introduced to select the action to be executed trying to balance exploration and exploitation. Analogously to Di (Nocera et al., 2012), in this paper we consider a Softmax method that selects the action to be executed through a Boltzmann distribution (Sutton and Barto, 1998):"

"Here, the temperature τ controls the exploration strategy: the higher the temperature, the closer the strategy is to a random policy (exploration); the lower the temperature, the closer the strategy is to Q(s, a) maximization (exploitation)."

(Zhang, 2024)

A Novel Multi-Objective Deep Q-Network: Addressing Immediate and Delayed Rewards in Multi-Objective Q-Learning

"However, Q-learning does not specify the exact nature of the exploration policy; alternatives like softmax [1], which selects actions based on a probability distribution derived from the action values, are also viable options. Actions are executed to observe rewards and new states, and Q-values are updated using the Bellman equation to reflect the sum of the observed reward and the discounted maximum future reward. This process repeats until the episode ends at a terminal state, a characteristic typical of episodic tasks or finite-horizon Markov Decision Processes (MDPs)."

(Higashi, 2025)

Individuality transfer: Predicting human decision-making across task conditions

"The probability of selecting action a was determined by a softmax policy:"

"where q it > 0 was a parameter called the inverse temperature or reward sensitivity, controlling the balance between exploration and exploitation"

".The Q-values are initialized as q init before an agent starts first episode."

(Moriyama et al., 2007)

Reinforcement Learning on a Futures Market Simulator

"For all s and a, Q t (s, a) is proved to converge to the true value under the optimal policy when (i) the environment has the Markov property, (ii) the agent visits all states and takes all actions infinitely, and (iii) α decreases properly [Watkins and Dayan 1992]. If the agent always chooses such actions during learning, however, Q t can converge to a local optimum because the agent may not visit all states. To avoid this, the agent typically uses a stochastic method like softmax [Sutton and Barto 1998] to choose actions. Softmax calculates action choice probabilities p s t (a), where a A(s t), as"

"T is called the temperature, with T > 0, and controls the effect of randomness."

(Diaz et al., 2025)

Global Convergence of Policy Gradient for Entropy Regularized Linear-Quadratic Control with multiplicative noise

"In the discrete action space setting, greedy policy (Watkins et al., 1992) and Botzman (Softmax) policy (Silver et al., 1998) are two effective and popular ways to balance exploration-exploitation and numerous developments have been made based on them."

(Zhang et al., 2019)

Global Convergence of Policy Gradient Methods to (Almost) Locally Optimal Policies

"However, this finite-horizon rollout, though generally used in practice, is known to introduce bias in estimating an infinite-horizon discounted value function. Such a bias in estimating the policy gradient for infinite-horizon problems has been identified in the earlier work (Baxter et al., 2001)[13], both analytically and empirically."

(Baxter et al., 2001)

Infinite-Horizon Policy-Gradient Estimation

"Gradient-based approaches to direct policy search in reinforcement learning have received much recent attention as a means to solve problems of partial observability and to avoid some of the problems associated with policy degradation in value-function methods. In this paper we introduce GPOMDP, a simulation-based algorithm for generating a biased estimate of the gradient of the average reward in Partially Observable Markov Decision Processes (POMDPs) controlled by parameterized stochastic policies. A similar algorithm was proposed by Kimura, Yamamura, and Kobayashi (1995). The algorithm's chief advantages are that it requires

storage of only twice the number of policy parameters, uses one free parameter β ∈ [0, 1] (which has a natural interpretation in terms of bias-variance trade-off), and requires no knowledge of the underlying state. We prove convergence of GPOMDP, and show how the correct choice of the parameter β is related to the mixing time of the controlled POMDP. We briefly describe extensions of GPOMDP to controlled Markov chains, continuous state, observation and control spaces, multiple-agents, higher-order derivatives, and a version for training stochastic policies with internal states. In a companion paper (Baxter, Bartlett, & Weaver, 2001) we show how the gradient estimates generated by GPOMDP can be used in both a traditional stochastic gradient algorithm and a conjugate-gradient procedure to find local optima of the average reward."

Effects of early random rewards and stochastic outcomes

In finite-horizon Q-learning with softmax, early lucky or unlucky rewards can push Q-values in the wrong direction for a long time. Because softmax uses those current Q-values to choose actions, early noise can change later sampling and make learning look unstable or slow. (3 sources)

A key finite-time issue is that early rewards are noisy, and Q-learning updates directly from those noisy outcomes. If one action happens to get a few lucky rewards early, its Q-value can rise enough that the agent keeps choosing it more often, even when it is not truly best. Barber gives a simple example where a high-variance reward on one action makes standard Q-learning spend a long time taking that action, because random positive outcomes make the agent think the bad action is good; only later does it slowly correct that mistake. Barber also notes that these early over-estimates can come from “unhelpful random rewards” that confuse the learner. (Barber, 2023)

With softmax, this matters because action choice is tied smoothly to the current Q-values rather than being fully random or fully greedy. So an early reward shock does not just affect one update; it also shifts future action probabilities, which changes what data the agent sees next. In stochastic environments, that can make reward trajectories very noisy in finite runs. Reported results show high-variance reward curves due both to environment stochasticity and to exploration under the softmax policy, with very low inverse temperature giving highly variable performance and very high inverse temperature sometimes causing early fixation on suboptimal actions and volatile rewards. (Sultana et al., 2026)

This is especially problematic when value differences between actions are small. Under Boltzmann or softmax Q-learning, actions with higher estimated Q-values get chosen more often, and the temperature schedule controls how quickly the policy stops exploring. Singh et al. show that when suboptimality gaps are small, regret for Boltzmann Q-learning can be arbitrarily close to linear, which matches the intuition that early random outcome differences can keep the learner spending many steps on the wrong actions before the signal becomes clear. (Singh et al., 2026)

So, in finite horizons, early random rewards do not just add harmless noise. They can redirect the learning path, change later sampling through softmax, and create long-lived errors or unstable reward curves even though asymptotically the method may still converge. (Barber, 2023) (Sultana et al., 2026) (Singh et al., 2026)

Evidence

(Barber, 2023)

Smoothed Q-learning

"This example is challenging for standard Q-learning since, because of the high variance in the reward, going Left from A will likely result in some positive rewards (whereas the optimal action is always to go Right from A). For this reason, Q-learning can spend a long time taking Left actions from A until it overcomes the stochastic nature of the reward"

". The example shows that double Q-learning avoids some of the issues with Q-learning and pays less attention to initial over-estimates of the Q value (due to unhelpful random rewards that confuse the model to thinking that going left is a good strategy)."

(Sultana et al., 2026)

A Multi-Model Adaptive Q-Learning Framework for Robust Portfolio Management in Stochastic Markets

"Reward curves have high variance in all environments due to the stochasticity of the environment and the exploration in the softmax policy"

". Specifically, β = 0.0 (i.e., random choice) led to highly variable performance, while extremely high β values (≥ 2.0) sometimes led to premature fixation on suboptimal actions, causing reward volatility."

(Singh et al., 2026)

Regret and Sample Complexity of Online Q-Learning via Concentration of Stochastic Approximation with Time-Inhomogeneous Markov Chains

"At each time step, actions with higher estimated Q-values are selected with higher probability, while suboptimal actions continue to be explored depending on the temperature parameter. This exploration strategy provides a smooth tradeoff between exploration and exploitation and has been widely used in reinforcement learning due to its simplicity"

".The parameter b controls the exploration-exploitation tradeoff, with larger values leading to faster decay of the temperature and hence reduced exploration"

".We first characterize how the regret of Boltzmann Q-learning depends on the suboptimality gap of the Markov decision process, showing that it can be arbitrarily close to linear for small gaps."

Sensitivity to initialization and early value estimates

Finite-horizon Q-learning with softmax is sensitive to how Q-values start out, because the policy uses those initial estimates before much real evidence is available. In short runs, bad or biased initial values can shape which actions get sampled, slow correction, and leave a lasting effect even though asymptotic convergence may still hold. (3 sources)

Another source of finite-time path sensitivity is the starting Q-table itself. In episodic Q-learning, the agent begins from initialized Q-values and chooses actions from those estimates before it has much data, so early behavior is partly driven by initialization rather than by the true action values. Both Rahmani et al. and Higashi state this setup directly: the Q-function is initialized before the first episode, and action choice is then based on the current Q-values under the exploration rule (Rahmani et al., 2020) (Higashi, 2025).

With softmax, this makes initialization especially important because even small early value differences are turned into different action probabilities. If one action starts with a higher estimated value, softmax will sample it more often, and that extra sampling can reinforce its lead before competing actions have enough observations to catch up (Higashi, 2025) (Model-Generated).

A common response is optimistic initialization: set Q-values high so the agent will try actions that have not yet been tested much. But this is a tradeoff, not a free fix. Sharma et al. note that higher initial Q-values can encourage exploration, yet they can also increase convergence time because the estimates may be poor and those poor estimates can persist for a long time during learning. They also point out that this kind of exploration happens mostly early, so it is not very helpful when uncertainty changes over time (Sharma et al., 2018).

So in finite horizons, initialization matters in two linked ways. First, it changes early softmax probabilities and therefore which data the learner collects. Second, if some state-action pairs are not experienced enough, the agent may never correct those starting beliefs within the available time, which is one reason finite runs can end far from the optimal policy even though infinite-experience convergence results exist in principle (Rahmani et al., 2020).

Evidence

(Rahmani et al., 2020)

Enhancing reinforcement learning by a finite reward response filter with a case study in intelligent structural control

"In the beginning the Q-function is randomly initialized and the agent selects actions based on its Q-function and the current state"

*".During the learning phase, the agent improves its estimations and after an infinite amount of experiences of all stateaction pairs the Q-values will converge to the optimal Q-values Q * (s, a) (Watkins et al., 1992)"*

".Otherwise, it will not experience all possible state-action values and most probably not learn the optimal policy."

(Higashi, 2025)

Individuality transfer: Predicting human decision-making across task conditions

> *"The probability of selecting action a was determined by a softmax policy:"*
>
> *"where q it > 0 was a parameter called the inverse temperature or reward sensitivity, controlling the balance between exploration and exploitation"*
>
> *".The Q-values are initialized as q init before an agent starts first episode."*

(Sharma et al., 2018)

Toward Massive Machine Type Communications in Ultra-Dense Cellular IoT Networks: Current Issues and Machine Learning-Assisted Solutions

> *"For example, τ = 0 case represents no exploration at all while T → ∞ case reflects the scenario in which the learning device chooses the action values almost with equal probability"*
>
> *".This approach encourages exploration by assigning higher initial values to the Q-function. However, the convergence time will be increased since the estimated Q-values can be quite bad estimates of the actual Q-value and these bad estimates may last longer during the learning process. Also, in the presence of dynamic uncertainty, this technique is not useful since the exploration mostly occurs at the beginning of the learning process."*

Path dependence from early misleading or adverse outcomes

In finite-horizon Q-learning with softmax, early bad luck can push learning onto a bad track. Once one action gets favored, softmax can keep returning to it often enough that the learner may not recover within the available time. (6 sources)

A central finite-time problem is path dependence: the final policy can depend heavily on the particular early trajectory, not just on the environment itself. If an action gets misleadingly good early outcomes, standard Q-learning can keep selecting it for a long time before enough contrary evidence arrives. Barber's example makes this concrete: high-variance rewards can make the learner persist with the wrong action because early random positives create an over-estimate that is slow to undo. (Barber, 2023)

Softmax does not remove this issue, because it uses the current Q-values to shape future action probabilities. Once learning enters an intermediate phase where temperature is lower and the currently favored action is chosen more often, other parts of the state space may only be visited partially, which can lock in a suboptimal value function and policy. In the zero-temperature limit this becomes greedy behavior, but the same effect can already matter well before that if probabilities become skewed enough. (Amin et al., 2021)

This is one way early adverse outcomes can have lasting effects. If one action gets unlucky early rewards, its Q-value can fall, softmax can reduce its sampling, and the learner may then fail to gather the evidence needed to repair that estimate. Recent work states this directly: finite-time under-exploration can combine with random early rewards to create path dependence and sensitivity to initial Q-values, and even though every action keeps positive probability under softmax, some actions may be sampled too rarely in practical finite time to correct early mistakes. (Tandon et al., 2025)

There is also a softer failure mode where softmax does not fully commit to one action, but still converges toward a poor practical policy. Hooshyar et al. note that as Q-factors get close to each other, softmax can assign almost identical probabilities to admissible actions, which in their experiments led to convergence to a far-from-optimal operating policy. That means path dependence is not only about hard lock-in; it can also show up as drifting into a weak action mix that never sharpens into the right behavior during the finite run. (Hooshyar et al., 2020)

More broadly, this helps explain why asymptotic convergence guarantees can miss what matters in a single finite learning run. Over many repeated episodes and trajectories, distinct early accidents may average out, but in a resource-limited finite horizon the agent only lives through one or a few actual trajectories, so those unique early events remain behaviorally important instead of being washed away by infinite-time averaging.

(Loxley, 2025) In practice, this can appear as reward volatility and even premature fixation on suboptimal actions when softmax is made too sharp, showing again that the learning path itself matters, not just the limiting theory. (Sultana et al., 2026)

Evidence

(Barber, 2023)

Smoothed Q-learning

> *"This example is challenging for standard Q-learning since, because of the high variance in the reward, going Left from A will likely result in some positive rewards (whereas the optimal action is always to go Right from A). For this reason, Q-learning can spend a long time taking Left actions from A until it overcomes the stochastic nature of the reward"*

> *". The example shows that double Q-learning avoids some of the issues with Q-learning and pays less attention to initial over-estimates of the Q value (due to unhelpful random rewards that confuse the model to thinking that going left is a good strategy)."*

(Amin et al., 2021)

A Survey of Exploration Methods in Reinforcement Learning

> *"In the early phase of learning, the agent should be able to try different actions in each state. Later in the intermediate learning phase, if the agent's target policy takes the control of the action selection process, it may lead to a partial visitation of the state space, and thus a sub-optimal policy and value function"*

> *".If T m decreases, the probability of generating the action with the highest expected reward a t increases, leading to a decrease in the probability of selecting other actions and hence a lower probability of exploring the environment. In the limit of zero temperature T m 0, the agent uses the target policy to select greedy actions in order to maximize the obtained rewards."*

(Tandon et al., 2025)

ES-C51: Expected Sarsa Based C51 Distributional Reinforcement Learning Algorithm

> *"Such finite-time under-exploration can interact with random rewards early in learning to produce behaviours such as path dependence and sensitivity to initial Q values, which may persist for an extended period in finite-horizon learning. Softmax is not immune to these issues. Once estimates start to favour one action, the probabilities can become so skewed that other actions, though never having zero probability, may be sampled too infrequently in practical finite time to correct early inaccuracies."*

(Hooshyar et al., 2020)

Aggregation–Decomposition-Based Multi-Agent Reinforcement Learning for Multi-Reservoir Operations Optimization

> *"As previously mentioned, the main advantage of the Softmax policy is to explore more at the beginning of learning while increasing exploitation as learning continues. This behavior is controlled using the values of Q-factors while converging to the steady-state situation, that is, the action with a greater Q-factor should have a higher chance to be chosen in every interaction. However, while applying this policy, it is observed that the values of all Q-factors become almost close to each other as they converge to the steady-state situation. In other words, Softmax might end up with almost identical probabilities for admissible actions. Therefore, this action-taking policy may lead to a poor performance through converging to a far-optimal operating policy, which our numerical experiments accomplished herein confirmed this result too."*

(Loxley, 2025)

An Optimal Policy for Learning Controllable Dynamics by Exploration

> *"This distinction is roughly due to the fact that in our case of resource-limited exploration we only have a finite time horizon to work over, and so non-stationary policies are natural to apply. While reinforcement learning tasks often involve (effectively) infinite horizons and therefore stationary policies become simpler to work with. For example, in the Q-Learning algorithm the optimal policy is derived from a Q-factor that does not depend explicitly on time. The reason is that to satisfy convergence requirements learning episodes are repeated many times over, generally with different initial conditions leading to different trajectories. The final Q-factor is therefore an average over many different trajectories; i.e., any distinct temporal properties of individual trajectories have been averaged out. On the other hand, by searching in policy space we work directly with individual trajectories, ultimately selecting the best one and retaining all of its unique properties. Several variants of Q-Learning achieve optimal exploration over an infinite horizon by using randomized policies (see, for example, (Icann et al., 1995)). However, a randomized policy does not produce repeatable results due to its non-determinism, and individual trajectories are again averaged over. This behaviour is unsatisfactory for understanding why a particular policy should work well, or how best to explore an unknown environment over a finite horizon."*

(Sultana et al., 2026)

A Multi-Model Adaptive Q-Learning Framework for Robust Portfolio Management in Stochastic Markets

"Reward curves have high variance in all environments due to the stochasticity of the environment and the exploration in the softmax policy"

". Specifically, β = 0.0 (i.e., random choice) led to highly variable performance, while extremely high β values (≥ 2.0) sometimes led to premature fixation on suboptimal actions, causing reward volatility."

Finite-time under-sampling despite positive softmax probabilities

Softmax gives every action a nonzero probability, but in a finite run that does not mean every action is sampled enough to matter. In practice, some actions can sit in the tiny tail of the distribution, get almost no updates, and remain effectively ignored. (6 sources)

A practical limit of softmax is the gap between "positive probability" and "meaningful sampling." In theory, Q-learning converges if every state-action pair keeps getting updated, but several authors stress that this condition is hard to satisfy in real finite learning because the agent may simply not explore enough of the space within the time available. (Grana et al., 2011) (Choshen et al., 2018)

Softmax helps by biasing action choice toward better-looking actions, but that same bias can leave weak-looking actions with probabilities so small that they are almost never chosen in a finite horizon. Yao et al. make this point sharply: even though softmax gives strictly positive probability to all actions, many can fall into the extreme tail, so under finite-sample training they have negligible effect on updates and are effectively omitted from optimization. (Yao et al., 2025)

This means "nonzero chance" is not enough to protect learning from under-sampling. Once current estimates start favoring one action, the distribution can become skewed enough that other actions are sampled too rarely in practical finite time to fix bad early estimates, even though their probability never reaches zero. Tandon et al. describe exactly this finite-time failure mode, linking it to persistent path dependence and sensitivity to initial Q-values. (Tandon et al., 2025)

Temperature choice affects how severe this becomes. Larger decay or sharper Boltzmann choice reduces exploration faster, and Singh et al. show that with small suboptimality gaps the regret of Boltzmann Q-learning can be arbitrarily close to linear, which is consistent with spending a long time not gathering enough evidence on the right actions. (Singh et al., 2026)

There is also a second practical issue: softmax is not directed toward the least-known actions, only toward actions with attractive current values. Choshen et al. argue that this means it can still waste exploration on already visited trajectories while failing to push strongly into unexplored ones. And Hooshyar et al. report another finite-run problem: as Q-values become close, softmax can assign nearly identical probabilities across actions and still settle into a poor operating policy. (Choshen et al., 2018) (Hooshyar et al., 2020)

So the practical answer is: yes, in finite-horizon Q-learning with softmax, actions can have positive probability yet be sampled so little that they are functionally absent from learning. That is one reason asymptotic convergence guarantees can look reassuring on paper while finite-time behavior remains fragile in actual runs. (Grana et al., 2011) (Yao et al., 2025) (Tandon et al., 2025)

Evidence

(Grana et al., 2011)

Cooperative Multi-Agent Reinforcement Learning for Multi-Component Robotic Systems: guidelines for future research

"In (Watkins et al., 2004) convergence of Q-learning in a stationary environment to an optimal policy with probability 1 is proved, as long as all state-actions pairs keep being updated. This is a theoretically sensible condition, but hard to meet in practice because an agent may not be able to explore sufficient space to guarantee convergence."

(Choshen et al., 2018)

DORA The Explorer: Directed Outreaching Reinforcement Action-Selection

*"Formally, to guarantee convergence to Q * , the Q-Learning algorithm must visit each state-action pair infinitely many times. A naive random walk exploration is sufficient for converging asymptotically. However, such random exploration has two major limitations when the learning process is finite. First, the agent would not utilize its current knowledge about the world to guide its exploration. For example, an action with a known disastrous outcome will be explored over and over again. Second, the agent would not be biased in favor of exploring unvisited trajectories more than the visited ones -hence "wasting" exploration resources on actions and trajectories which are already well known to it"*

". Another popular alternative, emphasizing the preference to learn about actions associated with higher rewards, is to draw actions from a Boltzmann Distribution (Softmax) over the learned Q values, regulated by a Temperature parameter. While such approaches lead to more informed exploration that is based on learning experience, they still fail to address the second issue, namely they are not directed (Thrun, 1992) towards gaining more knowledge, not biasing actions in the direction of unexplored trajectories."

(Yao et al., 2025)

The Debate on RLVR Reasoning Capability Boundary: Shrinkage, Expansion, or Both? A Two-Stage Dynamic View

"Although the Softmax function guarantees strictly positive probabilities for all actions, a significant number of these actions lie within the extreme tail of the probability distribution. As a result, under finite-sample training conditions, such actions exert negligible influence on parameter updates and are effectively omitted during optimization (that is, π l (v) ≈ 0 leads to Δz l v ≈ 0). Interestingly, the overall dynamic appears to unfold in two distinct stages. (1) Exploitation stage, corresponding to capability boundary shrinkage: initially, the model mainly samples already-explored high-reward and low-reward tokens, rarely selecting the potentially optimal one. Driven by positive advantage estimates, the probability of the high-reward token increases while the low-reward token decreases. The potentially optimal token's probability remains largely unchanged (or may even decrease, Remark 2), suggesting that over-exploitation in this stage may cause capability boundary shrinkage."

(Tandon et al., 2025)

ES-C51: Expected Sarsa Based C51 Distributional Reinforcement Learning Algorithm

"Such finite-time under-exploration can interact with random rewards early in learning to produce behaviours such as path dependence and sensitivity to initial Q values, which may persist for an extended period in finite-horizon learning. Softmax is not immune to these issues. Once estimates start to favour one action, the probabilities can become so skewed that other actions, though never having zero probability, may be sampled too infrequently in practical finite time to correct early inaccuracies."

(Singh et al., 2026)

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".The parameter b controls the exploration-exploitation tradeoff, with larger values leading to faster decay of the temperature and hence reduced exploration"

".We first characterize how the regret of Boltzmann Q-learning depends on the suboptimality gap of the Markov decision process, showing that it can be arbitrarily close to linear for small gaps."

(Hooshyar et al., 2020)

Aggregation–Decomposition-Based Multi-Agent Reinforcement Learning for Multi-Reservoir Operations Optimization

"As previously mentioned, the main advantage of the Softmax policy is to explore more at the beginning of learning while increasing exploitation as learning continues. This behavior is controlled using the values of Q-factors while converging to the steady-state situation, that is, the action with a greater Q-factor should have a higher chance to be chosen in every interaction. However, while applying this policy, it is observed that the values of all Q-factors become almost close to each other as they converge to the steady-state situation. In other words, Softmax might end up with almost identical probabilities for admissible actions. Therefore, this action-taking policy may lead to a poor performance through converging to a far-optimal operating policy, which our numerical experiments accomplished herein confirmed this result too."

Practical implications and limits of asymptotic convergence guarantees

The main practical lesson is that the usual convergence guarantee for Q-learning is an infinite-time result, not a promise about what will happen in one short run. In finite horizons with softmax, early noise, initialization, and skewed sampling can matter more than the asymptotic theory suggests. (7 sources)

Putting the pieces together, the standard convergence result for Q-learning is best read as a conditional

asymptotic statement: if the environment is Markov, learning rates are controlled properly, and every state-action pair keeps being visited and updated indefinitely, then Q-values converge to the optimal ones. That is exactly why stochastic action choice such as softmax is introduced in the first place: always following the current best-looking action can prevent enough visitation for convergence. But this guarantee does not say that a finite run will recover from bad early rewards, poor initialization, or early sampling imbalances. (Moriyama et al., 2007) (Grana et al., 2011) (Rahmani et al., 2020)

In practice, the hard part is not that the theorem is false, but that its assumptions are hard to realize with limited data and time. Grana et al. state this directly: requiring all state-action pairs to keep being updated is theoretically sensible, but hard to meet because the agent may not explore enough of the space. Rahmani et al. make the same finite-time point from the learning-process side: without experience over all state-action pairs, the agent will most likely fail to learn the optimal policy. So positive exploration in principle is weaker than adequate coverage in practice. (Grana et al., 2011) (Rahmani et al., 2020)

This is also why temperature scheduling is a practical risk. Early on, the agent should try many actions, but as temperature falls, the policy gives more control to the current value estimates. Amin et al. note that this can lead to only partial visitation of the state space and hence a suboptimal policy and value function; in the zero-temperature limit, behavior becomes greedy. So if early estimates were distorted by random rewards or unlucky outcomes, annealing softmax too quickly can freeze those distortions into the data-collection process itself. (Amin et al., 2021)

A second practical lesson is that softmax is only a value-biased exploration rule, not a knowledge-seeking one. Choshen et al. argue that even informed schemes like Boltzmann exploration still do not specifically push the agent toward unexplored trajectories, so exploration resources can be spent on already familiar parts of the space while important unknown parts remain weakly sampled. That helps explain why “every action has some chance” can still fail badly in finite time: the problem is not only randomness level, but whether exploration is directed to reduce uncertainty where it matters. (Choshen et al., 2018)

A related design tradeoff is that simple fixes can help one finite-time problem while worsening another. Sharma et al. note that high initial Q-values can encourage early exploration, but can also slow convergence because bad initial estimates may last longer, and this mostly helps only at the beginning. So practical tuning of initialization and temperature is not a technical detail; it directly shapes whether finite-time learning escapes early accidents or gets stuck with them. (Sharma et al., 2018)

More broadly, finite-horizon control often cares about the behavior of one actual trajectory or a small number of runs, while asymptotic Q-learning theory averages over many repeated episodes and trajectories. Loxley makes this contrast explicit: repeated learning over many trajectories can wash out distinct temporal features of individual runs, whereas finite-horizon decision making keeps those unique path properties behaviorally important. That is why asymptotic convergence can coexist with strong practical sensitivity to early random rewards, initialization, adverse first outcomes, and under-sampling in a single finite run. (Loxley, 2025)

So the practical answer to the user’s question is: in finite-horizon Q-learning with softmax, the method can still be heavily shaped by early randomness, initialization, and path-dependent sampling, even though asymptotic theory says convergence is possible under ideal infinite-exploration conditions. The theory is useful for understanding what can happen in the limit; it is much weaker as a guarantee about what will happen on the limited time scale that practitioners usually care about. (Moriyama et al., 2007) (Grana et al., 2011) (Amin et al., 2021)

Evidence

(Moriyama et al., 2007)

Reinforcement Learning on a Futures Market Simulator

"For all s and a, Q t (s, a) is proved to converge to the true value under the optimal policy when (i) the environment has the Markov property, (ii) the agent visits all states and takes all actions infinitely, and (iii) α decreases properly [Watkins and Dayan 1992]. If the agent always chooses such actions during learning, however, Q t can converge to a local optimum because the agent may not visit all states. To avoid this, the agent typically uses a stochastic method like softmax [Sutton and Barto 1998] to choose actions. Softmax calculates action choice probabilities p s t (a), where a A(s t), as"

"T is called the temperature, with T > 0, and controls the effect of randomness."

(Grana et al., 2011)

Cooperative Multi-Agent Reinforcement Learning for Multi-Component Robotic Systems: guidelines for future research

"In (Watkins et al., 2004) convergence of Q-learning in a stationary environment to an optimal policy with probability 1 is proved, as long as all state-actions pairs keep being updated. This is a theoretically sensible condition, but hard to meet in practice because an agent may not be able to explore sufficient space to guarantee convergence."

(Rahmani et al., 2020)

Enhancing reinforcement learning by a finite reward response filter with a case study in intelligent structural control

"In the beginning the Q-function is randomly initialized and the agent selects actions based on its Q-function and the current state"

*".During the learning phase, the agent improves its estimations and after an infinite amount of experiences of all stateaction pairs the Q-values will converge to the optimal Q-values Q * (s, a) (Watkins et al., 1992)"*

".Otherwise, it will not experience all possible state-action values and most probably not learn the optimal policy."

(Amin et al., 2021)

A Survey of Exploration Methods in Reinforcement Learning

"In the early phase of learning, the agent should be able to try different actions in each state. Later in the intermediate learning phase, if the agent's target policy takes the control of the action selection process, it may lead to a partial visitation of the state space, and thus a sub-optimal policy and value function"

".If T m decreases, the probability of generating the action with the highest expected reward a t increases, leading to a decrease in the probability of selecting other actions and hence a lower probability of exploring the environment. In the limit of zero temperature T m 0, the agent uses the target policy to select greedy actions in order to maximize the obtained rewards."

(Choshen et al., 2018)

DORA The Explorer: Directed Outreaching Reinforcement Action-Selection

*"Formally, to guarantee convergence to Q * , the Q-Learning algorithm must visit each state-action pair infinitely many times. A naive random walk exploration is sufficient for converging asymptotically. However, such random exploration has two major limitations when the learning process is finite. First, the agent would not utilize its current knowledge about the world to guide its exploration. For example, an action with a known disastrous outcome will be explored over and over again. Second, the agent would not be biased in favor of exploring unvisited trajectories more than the visited ones -hence "wasting" exploration resources on actions and trajectories which are already well known to it"*

". Another popular alternative, emphasizing the preference to learn about actions associated with higher rewards, is to draw actions from a Boltzmann Distribution (Softmax) over the learned Q values, regulated by a Temperature parameter. While such approaches lead to more informed exploration that is based on learning experience, they still fail to address the second issue, namely they are not directed (Thrun, 1992) towards gaining more knowledge, not biasing actions in the direction of unexplored trajectories."

(Sharma et al., 2018)

Toward Massive Machine Type Communications in Ultra-Dense Cellular IoT Networks: Current Issues and Machine Learning-Assisted Solutions

"For example, τ = 0 case represents no exploration at all while T ∞ case reflects the scenario in which the learning device chooses the action values almost with equal probability"

".This approach encourages exploration by assigning higher initial values to the Q-function. However, the convergence time will be increased since the estimated Q-values can be quite bad estimates of the actual Q-value and these bad estimates may last longer during the learning process. Also, in the presence of dynamic uncertainty, this technique is not useful since the exploration mostly occurs at the beginning of the learning process."

(Loxley, 2025)

An Optimal Policy for Learning Controllable Dynamics by Exploration

"This distinction is roughly due to the fact that in our case of resource-limited exploration we only have a finite time horizon to work over, and so non-stationary policies are natural to apply. While reinforcement learning tasks often involve (effectively) infinite horizons and therefore stationary policies become simpler to work with. For example, in the Q-Learning algorithm the optimal policy is derived from a Q-factor that does not depend explicitly on time. The reason is that to satisfy convergence requirements learning episodes are repeated many times over, generally with different initial conditions leading to different trajectories. The final Q-factor is therefore an average over many different trajectories; i.e., any distinct temporal properties of individual trajectories have been averaged out. On the other hand, by searching in policy space we work directly with individual trajectories, ultimately selecting the best one and retaining all of its unique properties. Several variants of Q-Learning achieve optimal exploration over an infinite horizon by using randomized policies (see, for example, (Icann et al., 1995)). However, a randomized policy does not produce repeatable results due to its non-determinism, and individual trajectories are again averaged over. This behaviour is unsatisfactory for understanding why a particular policy should work well, or how best to explore an unknown environment over a finite horizon."